3.4 - 3.4.2

Friday, April 10, 2026 11:34 AM

Exercise 25. Compare both expansions of P_7 from $\mathbb{Z}[x]$ in example 18 and from from $\mathbb{Z}_6[x]$ in example 19 , and consider the definition of $\mathbb{Z}_6$ as given in example 11. Can you see how the definition of P_7 over $\mathbb{Z}$ projects to the definition over $\mathbb{Z}_6$ if you consider the residue classes of $\mathbb{Z}_6$?

example 18

$P_7(x) = (x-2)(x+3)(x-5)$

$P_7(x) = x^3 - 4x^2 - 11x + 30$

example 19

$$
\begin{aligned}
(x-2)(x+3)(x-5) &= (x+4)(x+3)(x+1) && \text{\# additive inverses in } \mathbb{Z}_6 \\
&= (x^2+4x+3x+3\cdot 4)(x+1) && \text{\# bracket expansion} \\
&= (x^2+1x+0)(x+1) && \text{\# computation in } \mathbb{Z}_6 \\
&= x^3+x^2+x^2+x && \text{\# bracket expansion} \\
&= x^3+2x^2+x
\end{aligned}
$$

example 11

Example 11 (Arithmetic modulo 6). Choosing the modulus $n = 6$, we have six remainder classes of integers that are congruent modulo 6, that is, they have the same remainder when divided by 6. When we identify each of those remainder classes with the remainder, we get the following identification:

$$
\begin{aligned}
0 &:= \{\ldots,-6,0,6,12,\ldots\} \\
1 &:= \{\ldots,-5,1,7,13,\ldots\} \\
2 &:= \{\ldots,-4,2,8,14,\ldots\} \\
3 &:= \{\ldots,-3,3,9,15,\ldots\} \\
4 &:= \{\ldots,-2,4,10,16,\ldots\} \\
5 &:= \{\ldots,-1,5,11,17,\ldots\}
\end{aligned}
$$

take coef of $P_7(x) = x^3 - 4x^2 - 11x + 30$ and replace with remainder classes we get

$1 \to 1$

$-4 \to 2$

$-11 \to 1$

$30 \to 0$

$x^3 + 2x^2 + x$

which is exactly expansion in $\mathbb{Z}_6$

Exercise 26. Compare the sum $P+Q$ and the product $P \cdot Q$ from the previous two examples 22 and 23, and consider the definition of $\mathbb{Z}_6$ as given in example 11. How can we derive the computations in $\mathbb{Z}_6[x]$ from the computations in $\mathbb{Z}[x]$?

Example 22. To give an example of how polynomial arithmetic works, consider the following two integer polynomials $P, Q \in \mathbb{Z}[x]$ with $P(x) = 5x^2 - 4x + 2$ and $Q(x) = x^3 - 2x^2 + 5$. The sum of these two polynomials is computed by adding the coefficients of each term with equal exponent in x. This gives the following:

$$\begin{aligned}(P+Q)(x) &= (0+1)x^3 + (5-2)x^2 + (-4+0)x + (2+5) \\ &= x^3 + 3x^2 - 4x + 7\end{aligned}$$

The product of these two polynomials is computed by multiplying each term in the first factor with each term in the second factor. We get the following:

$$\begin{aligned}(P \cdot Q)(x) &= (5x^2 - 4x + 2) \cdot (x^3 - 2x^2 + 5) \\ &= (5x^5 - 10x^4 + 25x^2) + (-4x^4 + 8x^3 - 20x) + (2x^3 - 4x^2 + 10) \\ &= 5x^5 - 14x^4 + 10x^3 + 21x^2 - 20x + 10\end{aligned}$$

Example 23. Let us consider the polynomials of the previous example 22, but interpreted in modular 6 arithmetic. So we consider $P, Q \in \mathbb{Z}_6[x]$ again with $P(x) = 5x^2 - 4x + 2$ and $Q(x) = x^3 - 2x^2 + 5$. This time we get the following:

$$\begin{aligned}(P+Q)(x) &= (0+1)x^3 + (5-2)x^2 + (-4+0)x + (2+5) \\ &= (0+1)x^3 + (5+4)x^2 + (2+0)x + (2+5) \\ &= x^3 + 3x^2 + 2x + 1\end{aligned}$$

$$\begin{aligned}(P \cdot Q)(x) &= (5x^2 - 4x + 2) \cdot (x^3 - 2x^2 + 5) \\ &= (5x^2 + 2x + 2) \cdot (x^3 + 4x^2 + 5) \\ &= (5x^5 + 2x^4 + 1x^2) + (2x^4 + 2x^3 + 4x) + (2x^3 + 2x^2 + 4) \\ &= 5x^5 + 4x^4 + 4x^3 + 3x^2 + 4x + 4\end{aligned}$$

$(P+Q)(x) = x^3 + 3x^2 - 4x + 7$

$1 \to 1$

$3 \to 3$

$-4 \to 2$

$7 \to 1$

$= x^3 + 3x^2 + 2x + 1 \qquad \mathbb{Z}_6[x]$

$(P \cdot Q)(x) = 5x^5 - 14x^4 + 10x^3 + 21x^2 - 20x + 10$

$5 \to 5 \qquad 21 \to 3$

$-14 \to 4 \qquad -20 \to 4$

$10 \to 4 \qquad 10 \to 4$

$= 5x^5 + 4x^4 + 4x^3 + 3x^2 + 4x + 4$

Exercise 27. Consider the polynomial expressions $A(x) := -3x^4 + 4x^3 + 2x^2 + 4$ and $B(x) = x^2 - 4x + 2$. Compute the Euclidean Division of A by B in the following types:

1. $A, B \in \mathbb{Z}[x]$
2. $A, B \in \mathbb{Z}_6[x]$
3. $A, B \in \mathbb{Z}_5[x]$

Now consider the result in $\mathbb{Z}[x]$ and in $\mathbb{Z}_6[x]$. How can we compute the result in $\mathbb{Z}_6[x]$ from the result in $\mathbb{Z}[x]$?

Exercise 28. Show that the polynomial $B(x) = 2x^4 - 3x + 4 \in \mathbb{Z}_5[x]$ is a factor of the polynomial $A(x) = x^7 + 4x^6 + 4x^5 + x^3 + 2x^2 + 2x + 3 \in \mathbb{Z}_5[x]$, that is, show that $B|A$. What is A div B?

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$$
\begin{array}{r|l}
 & -3x^2 - 8x - 24 \\
\hline
x^2 - 4x + 2 & -3x^4 + 4x^3 + 2x^2 + 4 \\
 & -3x^4 + 12x^3 - 6x^2 \quad - \\
\hline
 & -8x^3 + 8x^2 + 4 \\
 & -8x^3 + 32x^2 - 16x \quad - \\
\hline
 & -24x^2 + 16x + 4 \\
 & -24x^2 + 96x - 48 \quad - \\
\hline
 & -80x + 52
\end{array}
$$

$\mathbb{Z}_6$

$$
\begin{array}{r|l}
 & 3x^2 + 4x \\
\hline
x^2 + 2x + 2 & 3x^4 + 4x^3 + 2x^2 + 4 \\
 & 3x^4 + 6x^3 + 6x^2 \quad - \\
\hline
 & 4x^3 + 2x^2 + 4 \\
 & 4x^3 + 2x^2 + 8x \quad - \\
\hline
 & 4x + 4
\end{array}
$$

$\mathbb{Z}$ to $\mathbb{Z}_6$

$-80 \to 4$

$52 \to 4$

$-3 \to 3$

$-8 \to 4$

$-24 \to 0$

$\mathbb{Z}_5$

$$
\begin{array}{r|l}
 & 2x^2 + 2x - 4 \\
\hline
x^2 + x + 2 & 2x^4 + 4x^3 + 2x^2 + 4
\end{array}
$$

$$
\begin{array}{r|l}
 & 2x^2+2x-4 \\
x^2+x+2 & 2x^4+4x^3+2x^2+4 \\
 & 2x^4+2x^3+4x^2 \quad - \\
\hline
 & 2x^3-2x^2+4 \\
 & 2x^3+2x^2+4x \quad - \\
\hline
 & -4x^2-4x+4 \\
 & -4x^2-4x-3 \quad - \\
\hline
 & 2
\end{array}
$$

$$
\begin{array}{r|l}
 & 3x^3+2x^2+2x+2 \\
2x^4-3x+4 & x^7+4x^6+4x^5+x^3+2x^2+2x+3 \\
 & x^7-4x^4+2x^3 \\
\hline
 & 4x^6+4x^5+4x^4+4x^3+2x^2+2x+3 \\
 & 4x^6+4x^3+3x^2 \quad - \\
\hline
 & 4x^5+4x^4+4x^2+2x+3 \\
 & 4x^5+4x^2+3x \quad - \\
\hline
 & 4x^4+4x+3 \\
 & 4x^4+4x+3 \quad - \\
\hline
 & 0
\end{array}
$$

$\leftarrow$ A div B

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